

STATISTICS 10 · IQR / BOX PLOTS

CLASS 2

FROM CONSTRUCTION TO INTERPRETATION

Fences · Outliers · Whiskers · Spread · Shape · Comparison

1

CLASSIFY

2

DRAW

3

READ

4

COMPARE

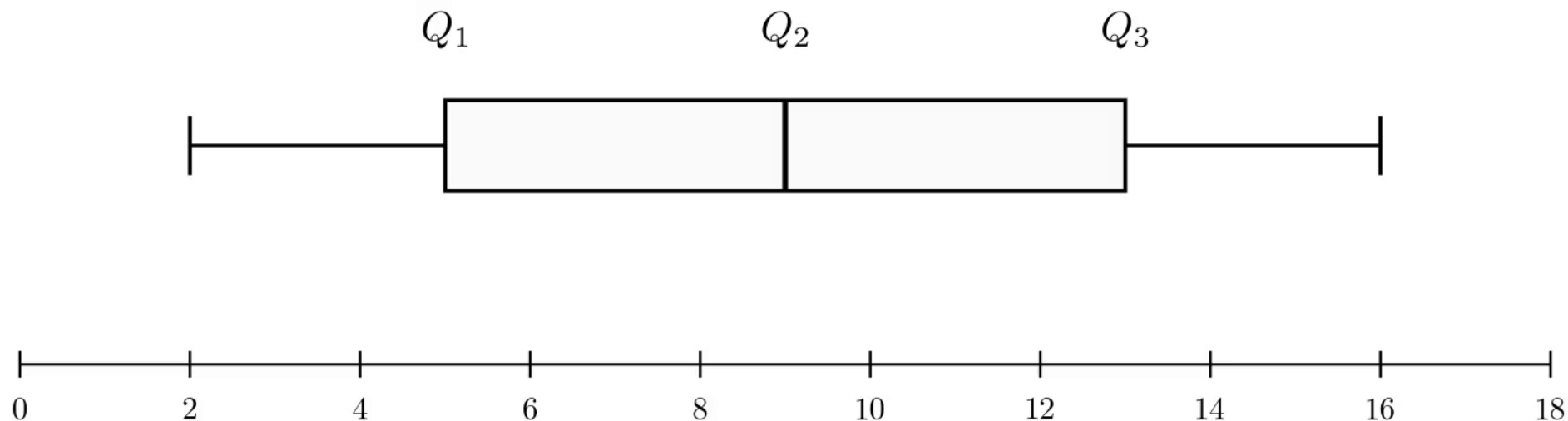
Last class we learned how to build the graph. Today we learn how to read it.

01

RECALL: WHAT DO WE ALREADY KNOW?

Read the box before adding a new rule: Q_2 describes center and $Q_3 - Q_1$ describes the middle 50%.

But what happens when one value is very far away?

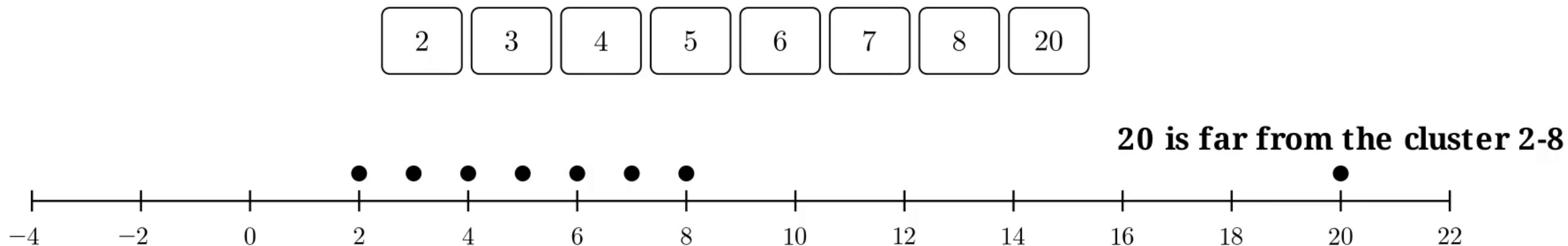


$Q_2 \rightarrow$ center

$Q_3 - Q_1 \rightarrow$ middle 50%

whiskers $\rightarrow$ extreme regular values

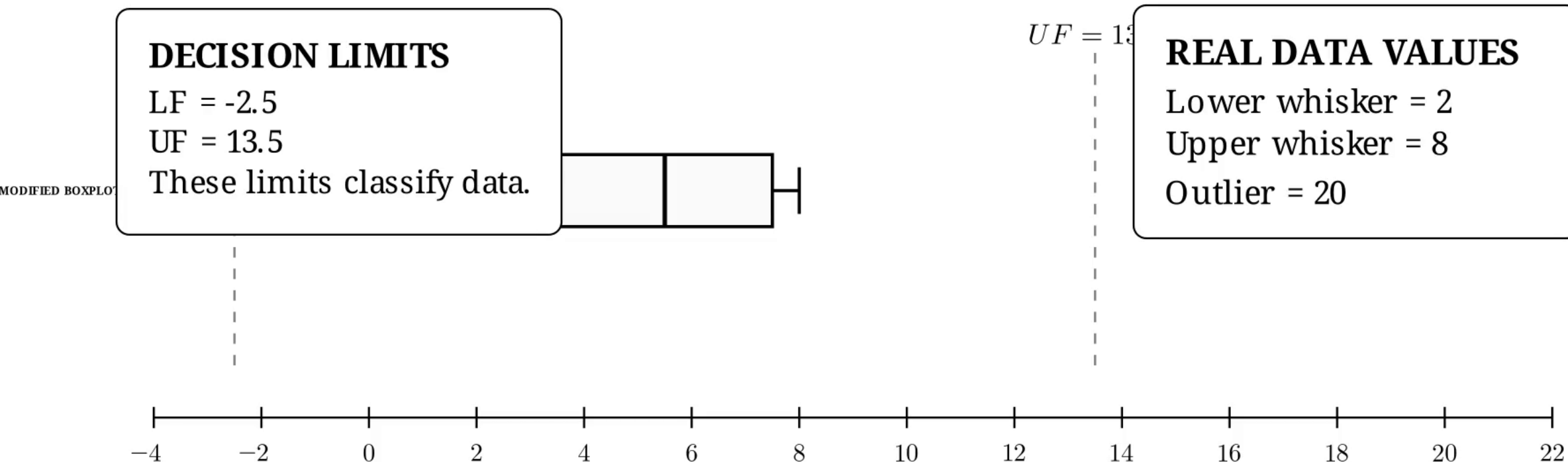
Fences are decision limits used to classify unusually distant observations; they are not data values we invent.



Should the whisker automatically extend all the way to 20?

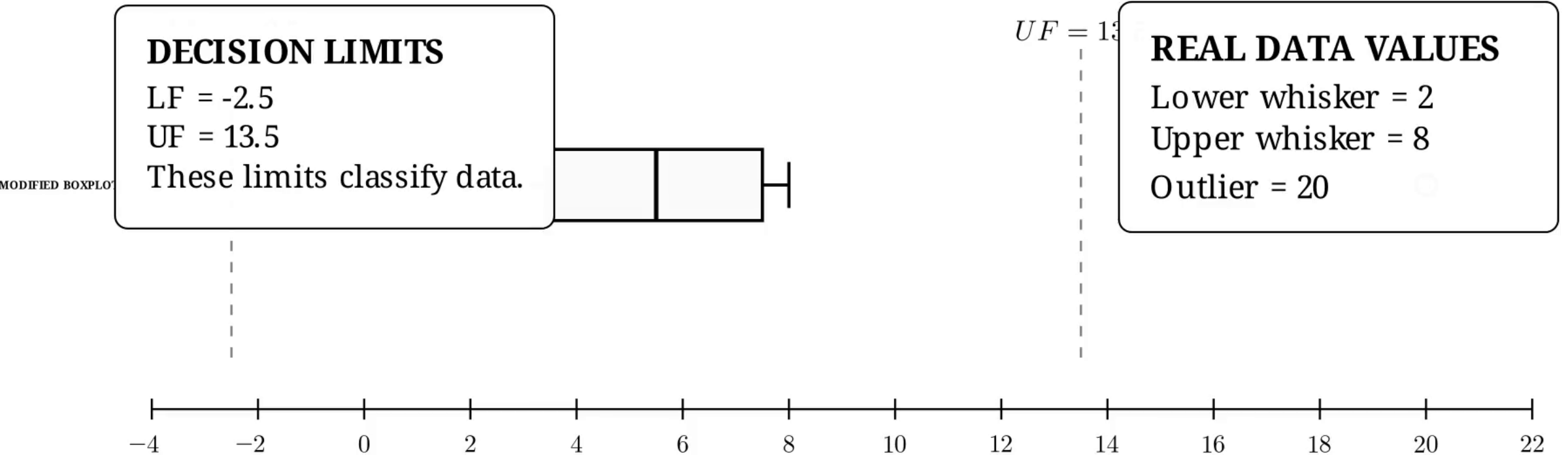
03 FENCES ARE NOT WHISKER ENDPOINTS

A modified boxplot stops each whisker at a real non-outlier observation and plots observations beyond the fences separately.



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whisker = most extreme REAL observation still inside the fences

WORKED EXAMPLE: CLASSIFY BEFORE DRAWING

The arithmetic creates the decision rule; the ordered data then determines the actual whisker endpoints.



$$IQR = 7.5 - 3.5 = 4$$

$$LF = 3.5 - 1.5(4) = -2.5$$

$$UF = 7.5 + 1.5(4) = 13.5$$

$$20 > 13.5 \Rightarrow 20 \text{ is an upper outlier}$$

NOW READ THE DATA

Smallest regular value: 2

Largest regular value: 8

20 is drawn as a separate point.

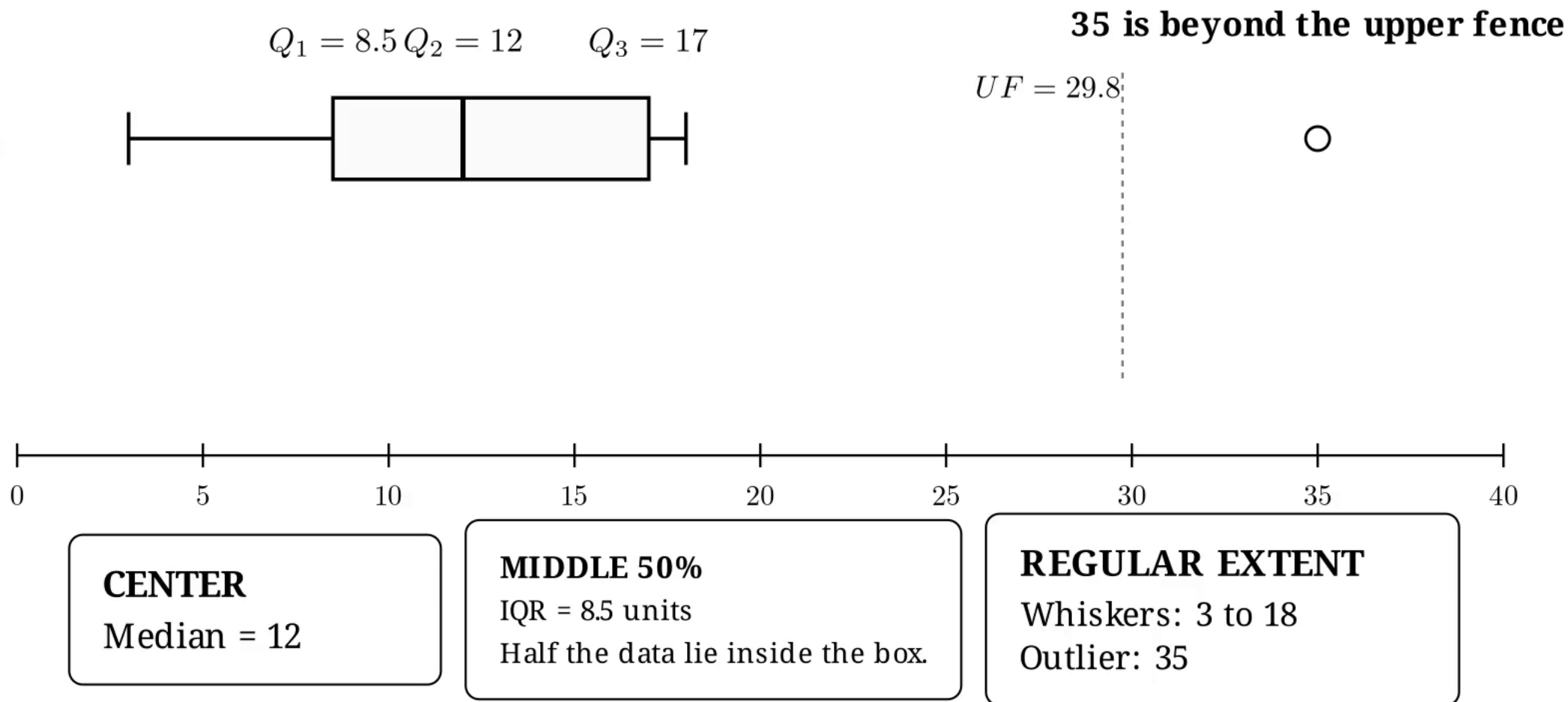
The whisker does NOT end at 13.5.'

05

READ THE GRAPH, NOT ONLY THE FORMULA

A boxplot communicates center, middle-50% spread, regular extremes and unusual observations on one numerical scale.

EXAMPLE



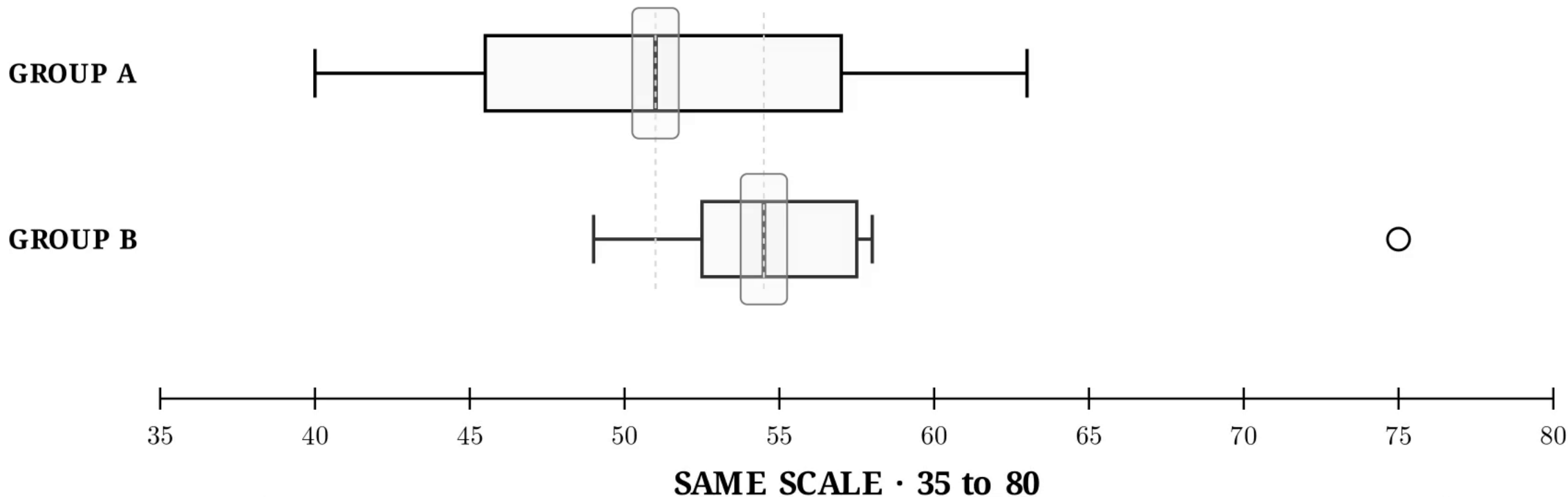
07

COMPARE CENTER FIRST

Median position answers a center question; do not confuse a higher median with greater variability.

A: 40, 44, 47, 50, 52, 55, 59, 63

B: 49, 52, 53, 54, 55, 57, 58, 75



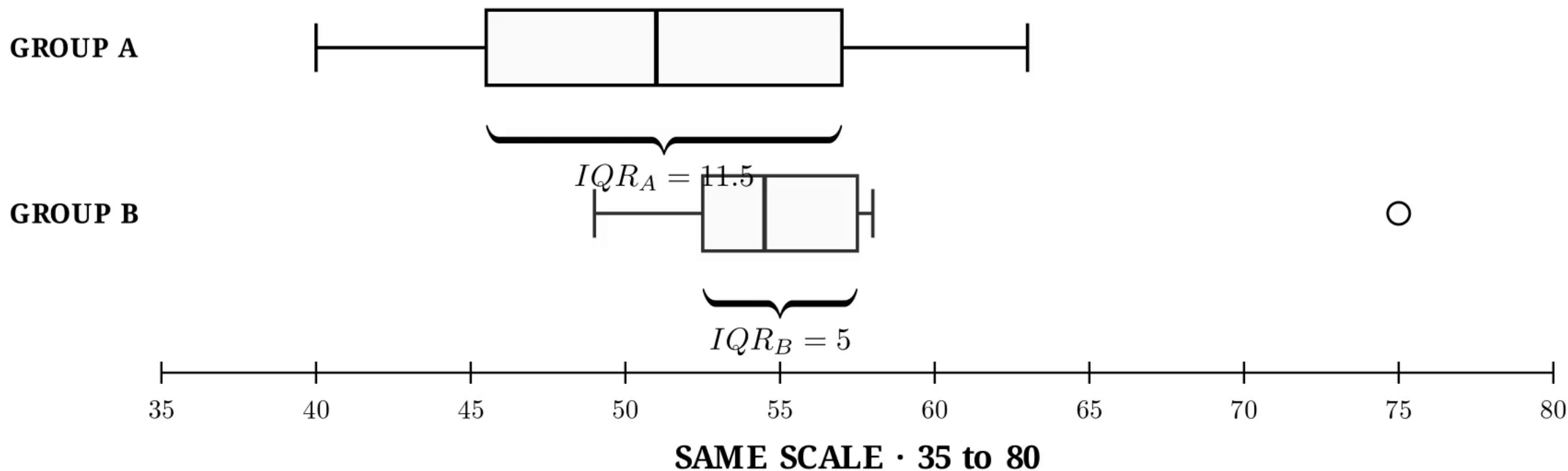
$Q_{2,B} = 54.5 > Q_{2,A} = 51 \Rightarrow$ Group B has the higher median.

08 COMPARE THE MIDDLE 50% WITH IQR

The box width is meaningful only because both boxplots use the same numerical scale.

A: 40, 44, 47, 50, 52, 55, 59, 63

B: 49, 52, 53, 54, 55, 57, 58, 75



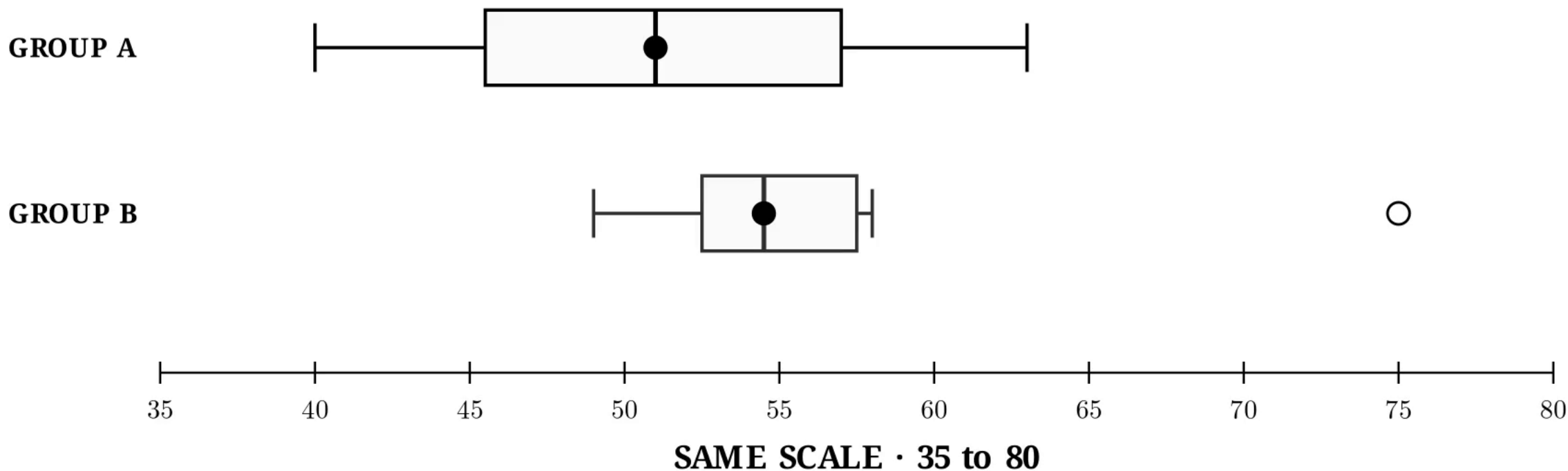
Group B has a smaller IQR, so its central observations are more tightly clustered.

SHAPE CLUES: BE CAUTIOUS

Unequal box halves or whiskers can suggest asymmetry, but a boxplot is a compact summary - not a complete picture of distribution shape.

Group A looks roughly balanced around its median.

Group B shows an upper-side irregularity - a clue, not proof of shape.



TURN NUMBERS INTO A STATISTICAL CONCLUSION

A strong comparison names the feature, cites evidence and explains what the evidence means in context.

CENTER

Group B has the higher median:
54.5 compared with 51.

MIDDLE 50%

Group B spans only 5 units.
Group A spans 11.5 units.

QUALIFICATION

Group B also contains
one unusually high value: 75.

Therefore: Group B is more tightly clustered in its central 50%, but it also has an unusually high observation.

COMMON MISTAKES TO AVOID

Each mistake changes the statistical meaning of the graph or makes a comparison invalid.

1

USING A FENCE AS A WHISKER

2

CONNECTING AN OUTLIER TO THE WHISKER

3

COMPARING PLOTS WITH DIFFERENT SCALES

4

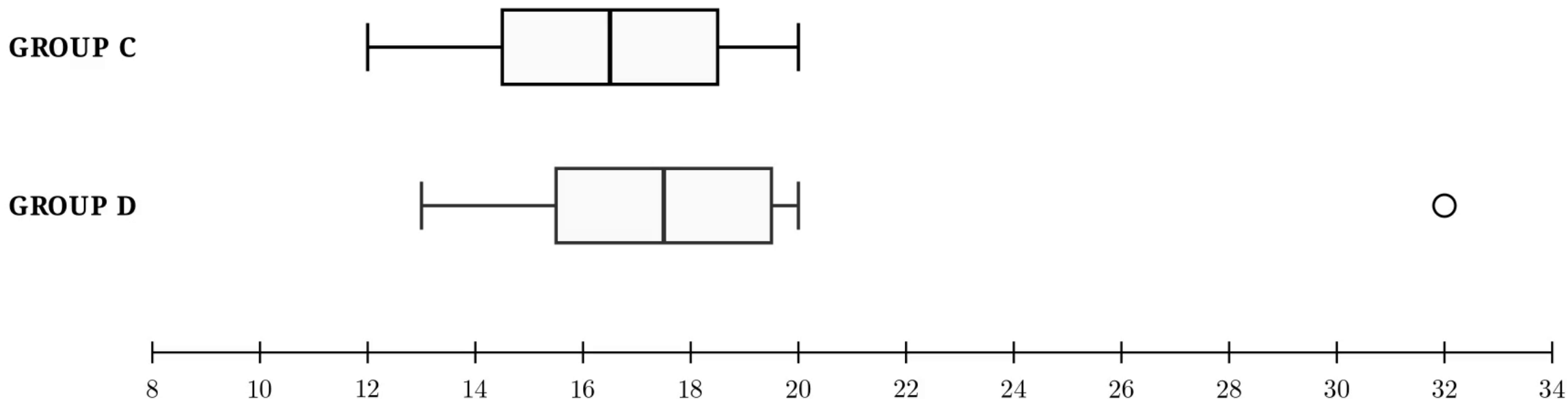
SAYING ONLY: 'THIS GRAPH IS BIGGER'

Compare: center + IQR + regular spread + asymmetry clues + outliers

YOUR TURN: COMPARE BEFORE THE REVEAL

Use the same order every time: center -> IQR -> whiskers
-> possible asymmetry -> outliers -> conclusion.

1) Which group has the higher median? 2) Compare IQR. 3) Identify any outlier.



$$Q_{2,D} = 17.5 > Q_{2,C} = 16.5$$

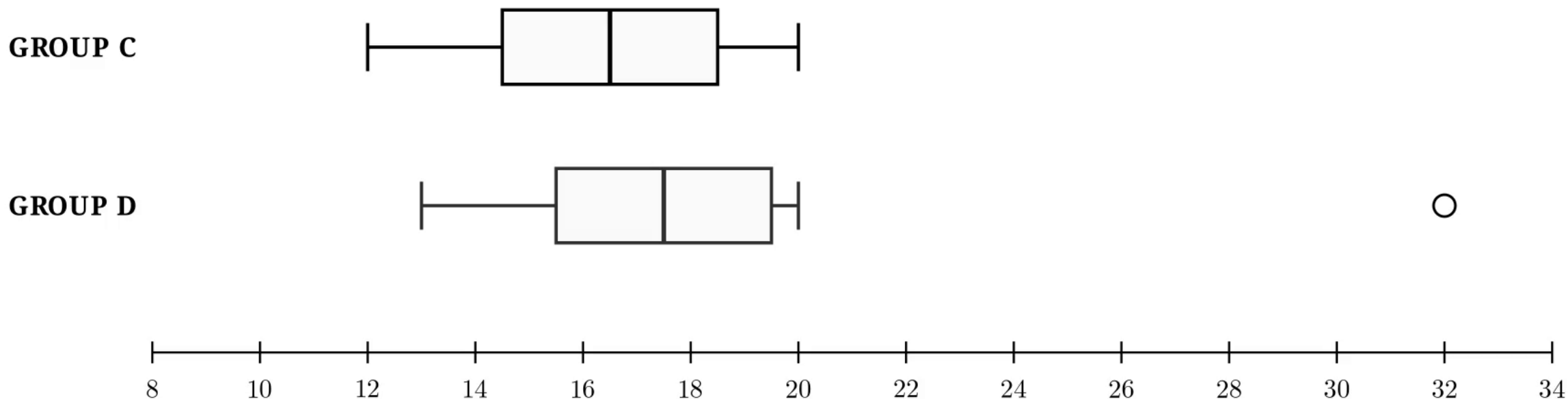
$$IQR_C = IQR_D = 4$$

$$32 > UF_D = 25.5 \Rightarrow \text{upper outlier}$$

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Today ends with interpretation. The next lesson generalizes position using deciles and percentiles.

1

QUARTILES

2

DECILES

3

PERCENTILES

$$Q_1 = P_{25}$$

$$Q_2 = P_{50}$$

$$Q_3 = P_{75}$$

Class 2 takeaway: construct correctly, then compare with evidence and precise statistical language.

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